

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introductions to Computer Architecture
In-Class Exercise (11/07)

1. Consider the following MIPS assembly program running on our standard **five-stage pipeline with forwarding to EX and RF bypassing**:

```
lw    $t0, 0($t5)
and   $t5, $t0, $t1
sub   $t2, $t3, $t5
lw    $t0, 4($t0)
sub   $t0, $t0, $t2
lw    $t2, 8($t3)
or    $t4, $t2, $t0
lw    $t2, 0($t0)
add   $t1, $t2, $t4
sw    $t3, 4($t1)
```

- a) How many **cycles** does it take to run the program above on our standard pipelined processor (from the IF stage of the first instruction to the WB stage of the last)?

Answer:

18 cycles = 14 cycles ideal + 4 cycles due to load-to-use stalls

- b) Reorder the assembly instructions above such that the program completes in **less cycles: 17 cycles**. Your reordered code should **not** change the program's output (i.e., none of the register data dependences should be violated). Do not add/remove/modify any instructions; **only reordering is allowed**.

Answer:

```
lw    $t0, 0($t5)
and   $t5, $t0, $t1
lw    $t0, 4($t0)
sub   $t2, $t3, $t5
sub   $t0, $t0, $t2
lw    $t2, 8($t3)
or    $t4, $t2, $t0
lw    $t2, 0($t0)
add   $t1, $t2, $t4
sw    $t3, 4($t1)
```

- c) You implement a physical register map table and free list for **register renaming**, initialized as follows:

Map Table	
Architectural Reg	Physical Reg
\$t0	P1
\$t1	P2
\$t2	P3
\$t3	P4
\$t4	P5
\$t5	P6

Free List	P7, P8, P9, P10, P11, P12, P13, P14, P15, P16, P17, P18, P20
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Apply register renaming (i.e., replace the architectural registers with physical registers) to the **original assembly program** above (not your reordered version), filling in the blanks below with physical registers. **Instructions must be renamed in program order.** Pop registers from the free list from left to right. You do not need to push registers back onto the free list. The first two instructions have been provided for you.

```

L0:  lw   P7, 0(P6)
L1:  and  P8, P7, P2
L2:  sub  _____, _____, _____
L3:  lw   _____, 4(_____)
L4:  sub  _____, _____, _____
L5:  lw   _____, 8(_____)
L6:  or   _____, _____, _____
L7:  lw   _____, 0(_____)
L8:  add  _____, _____, _____
L9:  sw   _____, 4(_____)

```

Answer:

```

L0:  lw   P7, 0(P6)
L1:  and  P8, P7, P2
L2:  sub  P9, P4, P8
L3:  lw   P10, 4(P7)
L4:  sub  P11, P10, P9
L5:  lw   P12, 8(P4)
L6:  or   P13, P12, P11
L7:  lw   P14, 0(P11)
L8:  add  P15, P14, P13
L9:  sw   P4, 4(P15)

```

- d) List all the **register data dependences** in the **renamed** program. For each one, specify the register, type of dependence (RAW, WAR, WAW) and pair of instructions. One dependence has been provided for you.

P7, RAW: L0 $\rightarrow$ L1

Answer:

P7, RAW: L0 $\rightarrow$ L3

P8, RAW: L1 $\rightarrow$ L2

P9, RAW: L2 $\rightarrow$ L4

P10, RAW: L3 $\rightarrow$ L4

P11, RAW: L4 $\rightarrow$ L6

P11, RAW: L4 $\rightarrow$ L7

P12, RAW: L5 $\rightarrow$ L6

P13, RAW: L6 $\rightarrow$ L8

P14, RAW: L7 $\rightarrow$ L8

P15, RAW: L8 $\rightarrow$ L9

- e) Reorder the **renamed** instructions such that the program now completes in the **minimum number of cycles possible: 14 cycles**. Your reordered code should **not** change the program's output (i.e., none of the register data dependences should be violated). Do not add/remove/modify any instructions; **only reordering is allowed**.

Answer:

```
lw    P7, 0(P6)
lw    P12, 8(P4)
and   P8, P7, P2
lw    P10, 4(P7)
sub   P9, P4, P8
sub   P11, P10, P9
lw    P14, 0(P11)
or    P13, P12, P11
add   P15, P14, P13
sw    P4, 4(P15)
```